

Part I: Theory & Definitions

(Chapter-wise Collection of Recurring Theory Questions)

Module 1: DC Circuits

1.1 Important Definitions & Laws

1. State Ohm's law and Kirchhoff's laws (KCL & KVL) in context with DC circuits. (S-25, W-19, W-22, S-22)
2. Define: Branch, Node, Junction, Mesh, Loop. (S-24, S-23)
3. Compare resistive series and parallel circuits. (S-25, W-23, S-19)

1.2 Theory Questions

1. **Star-Delta Transformation:** Derive the expression for equivalent resistances of a Star connected network to transform into a Delta connected network (and vice-versa). (S-25, W-24, W-23, S-24, W-19)
2. **Theorems:** State and explain Thevenin's Theorem, Norton's Theorem, and Superposition Theorem with suitable examples. (W-24, S-25, S-24, W-23, W-22, W-21)
3. **Circuit Analysis:** Apply KCL/KVL to solve simple circuits (e.g., Star connected circuit). (W-24, S-23)
4. **Transient Analysis:** Derive the expression for the voltage across the capacitor/current in inductor during charging (Time domain analysis). (W-24, W-21, S-22, W-23)

Module 2: AC Circuits

2.1 Important Definitions

1. Define: Amplitude, Frequency, Time Period, Instantaneous Value. (W-24, S-25, W-22, S-22)
2. Define: RMS Value, Average Value, Form Factor, Peak Factor. (S-25, W-24, S-24, S-22, W-19)
3. Define: Active Power, Reactive Power, Apparent Power, Power Factor. (S-25, W-24, W-23, S-22)
4. Define: Q-Factor, Bandwidth. (S-25, W-21)

2.2 Theory Questions

1. **Pure Circuits:** Prove that current in a purely Inductive/Capacitive circuit lags/leads voltage by 90° and average power consumption is zero. (S-25, W-23, S-24, S-22, S-20)
2. **Resonance:** Compare Series and Parallel Resonance. Write characteristics of Series Resonance. (W-24, S-25, W-22, S-22, W-19)
3. **Phasor Diagrams:** Draw Impedance, Voltage, and Power triangles for R-L/R-C series circuits. (S-25, S-22, W-21, W-24)
4. **Three Phase Relations:** Derive relationship between Line and Phase voltages/currents for Star and Delta connections. (S-25, W-24, W-23, S-24, S-22, W-19)
5. **Power Measurement:** Explain Two Wattmeter method for 3-phase power measurement (Phasor diagram, Merits/Demerits). (S-24, W-24, S-22, W-19)
6. **Polyphase Systems:** Advantages of 3-phase circuits over single-phase circuits. (S-25, S-23, W-21)

Module 3: Magnetic Circuits & Transformers

3.1 Important Definitions

1. Define: Magnetic Flux, Flux Density, MMF, Reluctance, Permeability, Field Strength. (W-24, S-25, S-23, W-19)
2. Define: Transformer, Regulation, Efficiency. (S-25)

3.2 Theory Questions

1. **Magnetic Circuits:** Explain B-H Curve and Hysteresis Loop. Explain Hysteresis and Eddy current losses. (W-24, S-25, S-23, S-20)
2. **Transformer Principle:** Explain working principle and construction of single-phase transformer. (S-25, W-22, S-23, W-19)
3. **EMF Equation:** Derive the EMF equation of a single-phase transformer. (S-25, W-24, S-24, W-23, W-22)
4. **Construction:** Compare Core type and Shell type transformers. (S-25, W-23, S-24, W-22)
5. **Auto Transformer:** Describe Auto Transformer with advantages, disadvantages, and applications. (S-25, W-24, S-24, W-23, W-22)
6. **3-Phase Transformers:** Explain various connections of 3-phase transformers (Star-Star, Delta-Delta, etc.). (S-25, W-23, S-20)
7. **Ideal vs Practical:** Comparison between ideal and practical transformer. (S-25)

Module 4: Fundamentals of Electrical Machines

4.1 Theory Questions

1. Induction Motor (1-Phase):

- Explain working principle. Why is it not self-starting? (S-25, S-23, S-20)
- Explain types: Split phase, Capacitor start/run (Construction & Working). (W-24, S-25, W-22, W-24)
- Merits, Demerits, and Applications. (S-25, W-24, W-22, W-19)

2. Induction Motor (3-Phase):

- Explain generation of Rotating Magnetic Field (RMF). (S-25, W-22, S-20, W-19)
- Compare Squirrel Cage vs. Slip Ring (Wound Rotor) Induction Motor. (S-25, W-24, W-23, S-22)

3. DC Machines:

- Explain construction and parts of DC Machine. (W-24, S-25, S-24, W-23)
- Classification of DC Motors. (S-25, S-24, W-23)
- Significance of Back EMF. (S-23, S-20)
- Brushless DC Motor (BLDC) construction and working. (W-24)

4. Synchronous Generator (Alternator): Explain construction and working principle. (S-25, W-24, W-23, S-22, S-19)

Module 5: Electrical Wiring, Safety & Protection

5.1 Theory Questions

1. Protective Devices:

- Explain MCB, ELCB, MCCB (Construction/Working). (W-24, S-25, W-23, S-20)
- Difference between Fuse and MCB. (S-25, W-23, W-22, W-21)
- Switch Fuse Unit (SFU). (S-22)

2. Earthing:

- Necessity of Earthing. Methods of Earthing. (S-25, W-24, S-24)
- Explain Plate Earthing and Pipe Earthing with diagrams. (W-24, S-25, W-23, W-18)

3. Safety:

- Safety precautions for electrical appliances/safety measures against shock. (W-24, S-25, W-23, S-23)
- List of safety devices for home appliances. (S-25, S-24)

4. Wiring & Cables:

- Structure of underground cable. (W-24, W-22, W-21)

- Difference between Wire and Cable. (S-23)
- 5. **Batteries:** Types, Characteristics, Construction of Lead Acid Battery. (W-24, S-25, S-23, S-20)
- 6. **Illumination:** Types of lamps, Lumen requirements. (W-24)
- 7. **Power Factor:** Methods of Power Factor improvement. Need for improvement. (S-24, W-24, W-23, W-22)
- 8. **Energy Meter:** Working principle of Energy Meter. (W-24)

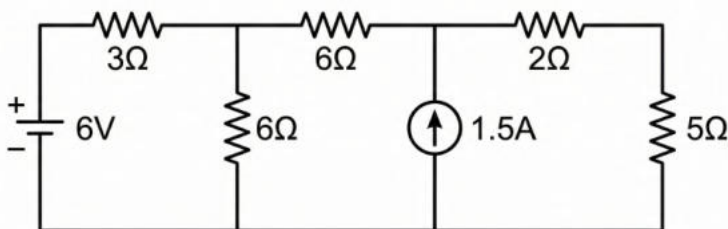
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Part II: Numerical Problems

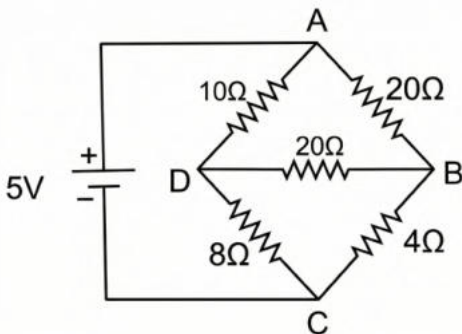
(Chapter-wise Collection of All Numericals from Past Papers)

Module 1: DC Circuits (Numericals)

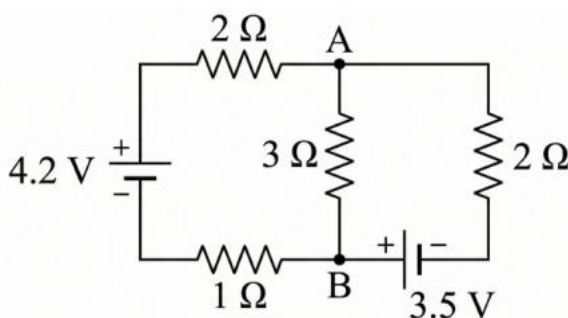
1. **(S-25) Superposition:** Find the current through 5Ω resistance using the Superposition theorem.



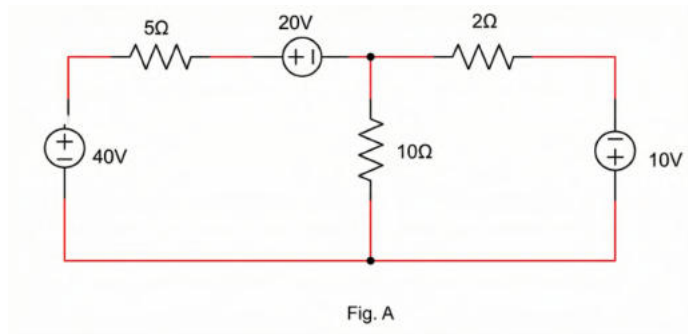
2. **(S-24) Thevenin's Theorem:** Using Thevenin's theorem, find current in branch BD of the network shown.



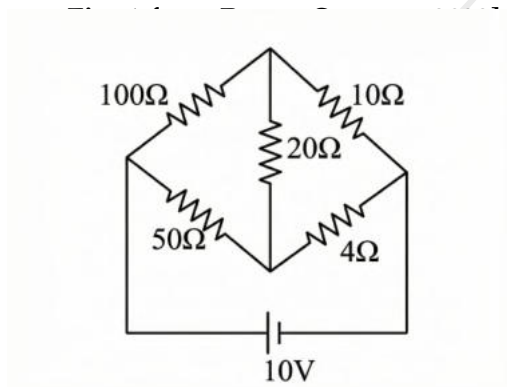
3. **(W-23) Superposition:** Use the superposition theorem to calculate the current in branch AB of the circuit shown.



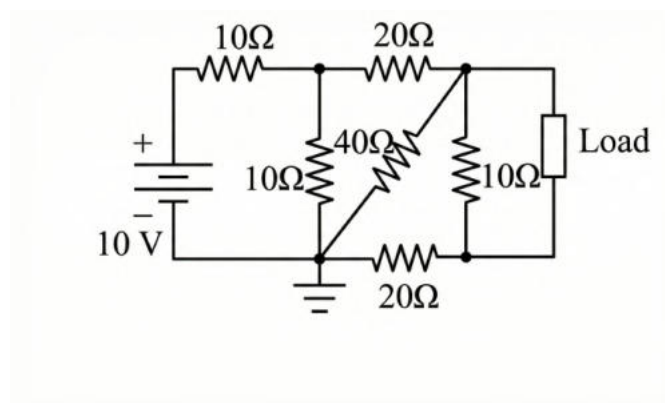
4. **(W-21) Thevenin's Theorem:** Find the current passing through 2Ω resistance using Thevenin's theorem.



5. **(S-24)** A 100 W, 230 V lamp is connected in series with a 50 W, 150 V lamp across 250 V supply mains. Calculate Voltage across each lamp.
6. **(W-22)** When three resistances of 10Ω , 20Ω , and 30Ω are connected in series across 230 V supply. Find (1) equivalent resistance (2) current flowing through each resistance, (3) voltage drop across each and (4) power loss in each resistor.
7. **(S-22)** For the Wheatstone bridge diagram shown, obtain the current flowing through the 20Ω resistance using Thevenin's equivalent network.

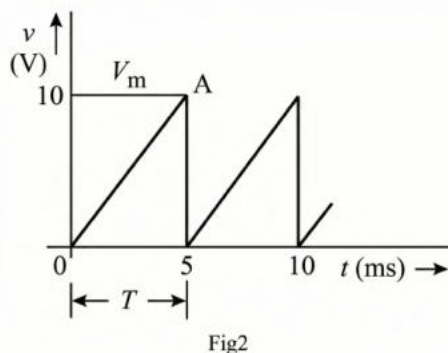


8. **(W-18) Norton's Theorem:** Obtain the value of Norton's equivalent current and resistance for the network shown.



Module 2: AC Circuits (Numericals)

1. **(S-23) Waveform Analysis:** Find the average and rms value for the waveform shown.



2. **(S-25)** A single phase R-L-C circuit having resistance of 8Ω , inductance of 80 mH and capacitance of $100\mu\text{F}$ is connected across single phase ac 150 V , 50 Hz supply. Calculate the current, power factor and voltage drop across inductance and capacitance.
3. **(S-25)** A 100 V , 100-Watt lamp is connected in series with a 100 V , 60-watt lamp across 200 V supply. Determine current drawn and power consumed by each lamp.
4. **(W-24)** If the waveform of a voltage has a form factor of 1.11 and peak factor of 1.51 and if the maximum value of a voltage is 4500 volts . Calculate the average and r.m.s. values of the voltage.
5. **(S-24)** A R-L-C series circuit consists of $R = 10\Omega$, $L = 0.1\text{H}$ and $C = 50\mu\text{F}$. Find 1. Impedance 2. Current and 3. Active power, when it is connected to a a.c. source of 230V , 50Hz .
6. **(S-24)** Three currents are represented by: $i_1 = 10\sin\omega t$, $i_2 = 20\sin(\omega t - \pi/6)$ and $i_3 = 30\sin(\omega t + \pi/4)$. Find magnitude and phase angle of resultant current.
7. **(W-23)** If the waveform of a voltage has a form factor of 1.15 and peak factor of 1.5 and if the maximum value of a voltage is 4500 volts . Calculate the average and r.m.s. values of the voltage.
8. **(S-23)** A 400 V , 50 Hz 3-phase supply has 100Ω between R and Y, $j100\Omega$ between Y and B and $-j100\Omega$ between B and R. Find (a) line currents for phase sequence RYB (b) star connected balanced resistors for the same power.
9. **(S-23)** A 120 V , 500 W lamp is used with a series choke on a 230 V , 60 Hz line, what is the inductance of the required choke if the choke has a Q of 2?
10. **(W-21)** A balanced 3-phase load consists of 3 coils each of resistance of 6Ω and inductive reactance of 8Ω . Determine line current and power absorbed when the coils are 1. Star connected and 2. Delta connected across 400V , 3-phase supply.
11. **(W-21)** A series resonance network consisting of a resistor of 30Ω , a capacitor of $2\mu\text{F}$ and an inductor of 20mH is connected across a sinusoidal supply voltage which has a constant output of 9 volts at all frequencies. Calculate the resonant frequency, current at resonance, voltage across inductor and capacitor at resonance, Quality factor, and bandwidth.

12. **(S-20)** A current of 5 A flows through a non-inductive resistance in series with a choking coil when supplied at 250V, 50Hz. If the voltage across the resistance is 125 V and across the coil 200V, calculate (i) impedance, reactance and resistance of the coil (ii) the power absorbed by the coil (iii) and total power.
13. **(W-19)** A single phase R-L-C circuit having resistance of 8Ω , inductance of 80mH and capacitance of $100\mu F$ is connected across single phase ac 150 V, 50Hz supply. Calculate the current, power factor and voltage drop across inductance and capacitance.
14. **(S-19)** A balanced star connected load of $(4 + j3)\Omega$ per phase is connected to a balance 3 phase 400 V supply. Find the line current, power factor, active power and reactive power.
15. **(S-19)** A series RLC circuit with $L=160$ mH, $C = 100\mu F$ and $R = 40\Omega$ is connected to a sinusoidal voltage $V(t) = 40 \sin \omega t$, with $\omega = 200$ rad/sec. Find (i) Impedance (ii) I_0 (iii) Phase.
16. **(W-18)** The maximum values of voltage and current in a circuit are 400 V and 20 A respectively. Both are sinusoidal with 50 Hz frequency. Instantaneous values at $t = 0$ are 283 V and 10 A. Obtain equations of voltage and current. Also find active power.
17. **(W-18)** In a series R-L circuit, a voltage of 10 V at 50 Hz frequency produces a current of 750 mA. In the same circuit with same magnitude of applied voltage with a frequency of 75 Hz produces a current of 500 mA. Find out the values of R and L.

Module 3: Magnetic Circuits & Transformers (Numericals)

Note: Few numericals appeared directly for this module in recent papers, focusing mostly on derivation and theory.

Module 4: Fundamentals of Electrical Machines (Numericals)

1. **(S-20)** A d-c generator has an e.m.f of 200 volts and provides a current of 10 amps. How much energy does it provide each minute?

Module 5: Electrical Wiring, Safety & Protection (Numericals)

1. **(S-25)** Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h, a water pump of 0.5 HP for 15 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watts.
2. **(W-24)** Compute the monthly energy charges for an air conditioner having Power consumption of 3 kW and daily uses 10 hours. Energy charges Rs. 5 per unit.
3. **(S-24)** Calculate the electricity bill amount for a month of March, if 5 bulbs of 40 W for 5 h, 6 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 2 h, a water pump of 0.5 HP for 30 minutes are used per day. The cost per unit is Rs 3.50.
4. **(W-23)** Compute the energy charges for an air conditioner having consumption of 2 kW for the month of April. Daily usage of the air conditioner is 12 hours. Energy charges are Rs 9 per unit.

5. **(W-22)** Compute the monthly energy charges for an air conditioner having Power consumption of 3 kW and daily uses 10 hours. Energy charges Rs. 5 per unit.
6. **(S-22)** Calculate the Resistance of a 100 m length of wire having a uniform cross sectional area of 0.02mm^2 and having resistivity of $40\mu\Omega\text{-cm}$.
7. **(W-21)** Calculate current, resistance and energy consumed by an oven rated 230V, 1KW when used for 20 hours. Also calculate the electricity bill at the rate of Rs. 7/- per unit.
8. **(S-20)** Calculate the resistance of a 100 m length of wire having a uniform cross sectional area of 0.02mm^2 and having resistivity of $40\mu\Omega\text{-cm}$.
9. **(S-19)** Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h, a water pump of 0.5 HP for 15 minutes are used per day. The cost per unit is Rs 3.50.
10. **(W-18)** Compute the monthly energy charges for an air conditioner having consumption of 2 kW. Daily usage of the air conditioner is 10 hours. Energy charges are Rs 8 per unit.

^^ End Of The Question Bank ^^



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